

Using Contemporary Constraints to Ensure Data Consistency

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ABSTRACT

Concern about the propagation of fake news and data grow every day. At the same time, the analysis and mining of large data sets have become essential in the decision-making processes, in which incorrect data bring misleading insights. However, there is no general-purpose, automatic mechanism to effectively ensure the consistency of temporal data in Linked Data sources such as Wikidata, Wikimedia's free knowledge base.

In this paper, an approach based on cross-comparing date values to discover inconsistent data is proposed. Besides, the concept of contemporary constraint, on which this approach is based, is defined and formalized in order to show how to find inconsistencies in a wide range of data sources. Our experimental results show that contemporary constraints are effective and can be used with multiple purposes for data curation and data quality analysis. As a success story, the contemporary constraint has been implemented in Wikidata.

CCS CONCEPTS

• **Information systems** → **Integrity checking**; *Data cleaning*;

KEYWORDS

data quality, temporal constraints, Linked Open Data, Wikidata, DBpedia

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1 INTRODUCTION

If someone claims: 1) that the Commonwealth of England and the Kingdom of Poland were at war, 2) that the Commonwealth of England was a brief republic restricted to the period between 1649 and 1660 and 3) that the Kingdom of Poland was dissolved or abolished in 1569, then we can conclude that this person is not providing reliable data, since coexisting at some point in history is an essential requirement to be at war. So, it can be concluded that:

- the Commonwealth of England and the Kingdom of Poland were never at war, or that
- the Commonwealth of England existed before 1649, or that
- the Kingdom of Poland was dissolved or abolished after 1569, or that
- several of these circumstances are true.

This example shows a type of temporal constraints which do not only affect countries at war, but also teachers and their students, companies and their directors, athletes and their teams, politicians and their parties, interconnected roads and streets, factories and their products, spouses, joining rivers, etc. All these kinds of relationships require that the same rules are respected. Therefore, analogous consistency checks and constraints are useful for ensuring the quality of the corresponding data sources.

Thanks to the technological improvements over last decades, high volumes of digital data have been generated. The exploitation of these data from diverse perspectives, such as the Semantic Web and data mining and science, is influenced by data quality, so measuring and ensuring the consistency of temporal data (i.e., data that consist essentially of dates and times) in an efficient way is key. On the other hand, temporal data are not usually considered explicitly in the axioms and rules of inference of conventional deductive systems [3]. So, inconsistency detection tools, such as the reasoners Pellet¹ and FaCT++², commonly used in the context of the Linked Data Web and the Semantic Web, do not exploit temporal data in depth. Furthermore, freely-editable knowledge bases tend to contain inconsistent data [10] due to the broad range of knowledge they store and the diversity of their editors' criteria. As well, many people are able to insert new data about a particular subject and each editor may be or may be not an expert in that subject. To minimize the impact of these inconsistencies, some constraints are usually introduced to help users.

In this paper, the concept of contemporary constraint is formalized and analyzed in order to develop a general-purpose, automatic mechanism to ensure the consistency of temporal data in the context of the Semantic Web and Linked Open Data sources. As a first success story based on this paper, the contemporary constraint has been implemented in Wikidata; after being approved by Wikidata's development team and its community, the constraint is actively detecting and monitoring inconsistent data in this freely-editable knowledge base since September, 2018.³ The contemporary constraint is the most sophisticated constraint in Wikidata so far and the only one which is cross-comparing quantitative data by using relationships between entities.

This paper is structured as follows. In Section 2, the contemporary constraint is formally defined and studied. Then, in Section 3,

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¹<http://pellet.owld.com/>

²<http://owl.man.ac.uk/factplusplus/>

³<https://grafana.wikimedia.org/dashboard/db/wikidata-quality>

a metric to determine the usefulness of using contemporary constraints is defined. After that, in Section 4, two methods to estimate both theoretically and practically the defined metric are presented and analyzed by studying multiple data sources; among them, Wiki-data [20] and DBpedia [2], two general knowledge bases widely used by the Semantic Web and Linked Open Data communities. Finally, related work, and conclusions and future work are depicted in Sections 5 and 6, respectively.

2 PROBLEM STATEMENT

Since we are dealing with temporal data, one possible way to formalize our problem is by means of a temporal logic. There is abundant literature about temporal logics [9]. Nevertheless, the definitions and properties presented in this section are general enough and do not require the use of logic formalisms, which would introduce unnecessary complexity to solve practical cases on real systems in the context of the Linked Open Data Web and the Semantic Web. Besides, since our approach is based on statistical properties, in this paper, the problem is formalized from an analytical point of view instead of a logical one. On the other hand, our definitions can easily be translated to other representation systems; in particular, in Allen's interval algebra [1], one of the most used formalisms for temporal reasoning [13]; as the induced contemporary relation, defined and studied in the next subsections, is equivalent to the union of all but two —precedes and preceded by— of the Allen's relations.

2.1 Definition of Contemporary

From now on, the extended real number set, $\mathbb{R} \cup \{-\infty, +\infty\}$, is used to represent time, establishing an order where low values represent older moments and high values represent later moments. $-\infty$ and $+\infty$ are reserved for undefined, unknown or non-applicable values. The term 'entity' refers everything with objective or conceptual existence in the universe. For instance, a metronome, the European Union, Hypatia of Alexandria, humanity, globalization and Homer Simpson are entities.

Definition 2.1. Let e be an entity, the start time of e , denoted by e_{min} , is defined as the moment to which it is said that e does not exist. Analogously, the end time of e , denoted by e_{max} , is defined as the moment from which it is said that e does not exist. As a consequence, the life interval of e , denoted by $\mathcal{I}_e$, is defined as $\mathcal{I}_e := [e_{min}, e_{max}]$.

In this context, two entities are considered *contemporary* if their life intervals intersect. Let us note that only one start and one end moments are considered for each entity. So, it is considered that, once an entity no longer exists, it cannot return to exist, that is, its identity is linked to its temporal continuity.

Definition 2.2. Let E be a set of entities. It is said that the entities $e_1, e_2 \in E$ are *contemporary*, and it is denoted by $(e_1, e_2) \in \mathcal{C}_E$, if, and only if, their life intervals hold that $\mathcal{I}_{e_1} \cap \mathcal{I}_{e_2} \neq \emptyset$. Note that $\mathcal{C}_E$ defines a relation between entities so that, $\mathcal{C}_E \subseteq E \times E$, $\mathcal{C}_E := \{(e_1, e_2) \mid \mathcal{I}_{e_1} \cap \mathcal{I}_{e_2} \neq \emptyset\}$. We say that $\mathcal{C}_E$ is the *induced contemporary relation* from the set of entities E .

Any relation R_E between entities that require coexistence at some point must satisfy the *contemporary constraint*, i.e., $R_E \subseteq \mathcal{C}_E$.

So, this constraint restricts the possible entities in E to which a specific entity $e \in E$ can be related. Contemporary constraints can be used together with range constraints for quantitative data (applied for dates), as they complement each other. In a relation where the range of acceptable start and end values for the entities is very wide, it is more probable that inconsistent data can be detected by the contemporary constraint than by the range constraint. On the other hand, if the range of acceptable values is narrow, then the probability that an element belongs to that range will be lower, and the range constraint will be more useful for detecting data inconsistencies than the contemporary constraint.

In the following, the features of contemporary relationships and the violations of the contemporary constraint are formalized and analyzed.

2.2 Contemporary Relationships

When modeling knowledge, relationships between entities are often introduced according to the relative function that every entity plays with respect to the other one or to the way in which both entities interact. It must be noted that certain relationships, because of their own definition or their physical limits in the real world, which are imposed by the associated semantics, can only be established between contemporary entities. Let us denote by $\mathcal{R}_C$ the set of all these relations.

Definition 2.3. Let E be a set of entities and R_E be a relationship where $R_E \subseteq E \times E$. R_E is said to be a *contemporary relationship* if, for any $(e_1, e_2) \in R_E$, it should hold that the corresponding life intervals of both entities intersect, $\mathcal{I}_{e_1} \cap \mathcal{I}_{e_2} \neq \emptyset$. We denote by $\mathcal{R}_C$ the set of relationships that, according to their semantics, must hold this condition for any E .

Notice that the relationships belonging to $\mathcal{R}_C$ can be used to detect temporal inconsistencies in a data source. If R_E is in $\mathcal{R}_C$, then R_E must be a subrelation of the induced contemporary relation $\mathcal{C}_E$, i.e., $R_E \subseteq \mathcal{C}_E$. On the other hand, if R_E is not a subset of $\mathcal{R}_C$, there exists inconsistent data in the studied data source. For example, *IsSpouseOf* is a contemporary relationship; if it is stated that the entity *Alice* is the spouse of the entity *Bob* but the life intervals of *Alice* and *Bob* do not meet, i.e., $\mathcal{I}_{e_{Alice}} \cap \mathcal{I}_{e_{Bob}} = \emptyset$, there exists an inconsistency.

In order to determine whether R_E is a contemporary relationship (i.e., R_E is in $\mathcal{R}_C$), its associated semantics must be studied, and not the particular data used to populate the data source. Thus, if, because of its semantics, R_E can be included in one or more of the following categories, then R_E will be a contemporary relationship:

- If R_E requires some synchronous communication, use or direct physical interaction between the linked entities, then R_E is a contemporary relationship. For example, *HasHeadCoach*, *Supervises*, *Establishes*, *Uses*, *FeedsOn*, *AtWarWith*, *MascotOf*, *Employer*, *PositionHeld*, *Manages*, *Finances*, etc. are contemporary relationships belonging to this category.
- If R_E describes the relative spatial location between the linked entities, then R_E is a contemporary relationship. For example, *NextTo*, *LocatedIn*, *SharesBorderWith*, *Crosses*, *EndemicTo*, *Contains*, *LivesIn*, *SurroundedBy*, *ConnectsWith*, etc. are contemporary relationships belonging to this category.

This belonging is due to the fact that, when comparing locations through a single relationship, time must be assumed to be the same for all the compared entities so that this relationship can provide meaningful information, and vice versa.

Summing up, we are dealing with a data source, or any information system, $\mathcal{D} = (E, \mathcal{I}_E, \mathcal{R}_E)$, where E is a set of entities, $E = \{e_1, \dots, e_n\}$, $\mathcal{I}_E$ is the set of the corresponding life intervals, $\mathcal{I}_E = \{\mathcal{I}_{e_1}, \dots, \mathcal{I}_{e_n}\}$, and $\mathcal{R}_E$ is a set of relationships that we know a priori that are contemporary, $\mathcal{R}_E \subseteq \mathcal{R}_C$.

Now, a question arises: is it possible to evaluate $\mathcal{D}$ to measure its capacity to detect temporal inconsistencies? In other words, in case that a temporal inconsistency existed in $\mathcal{D}$, would it be easy to detect it or not? In the following, this question is answered by studying the distribution of the induced contemporary relation, $\mathcal{C}_E$. But, firstly, we formally define what a contemporary constraint violation is.

2.3 Contemporary Constraint Violation

Having properly identified and labeled contemporary relationships, a software system can automatically detect when two entities that are linked through some of these relationships, according to the available information, are not contemporary. This situation indicates that wrong data are being stored in the data source.

Given a relationship R_E in $\mathcal{R}_C$, that is, R_E has been said contemporary, it is said that a *violation* v of the *contemporary constraint* exists if, and only if, $v \in R_E$ but $v \notin \mathcal{C}_E$.

If a relationship R_E has been properly identified as contemporary, every violation $v \in R_E \setminus \mathcal{C}_E$ such that $v = (e_1, e_2)$ will reveal that, at least, one of the following situations occurs:

- It is not true that e_1 and e_2 are related through R_E in the real world, that is to say, $(e_1, e_2) \notin R_E$.
- Value $\min\{e_{1max}, e_{2max}\}$ is wrong. This value should be later in time.
- Value $\max\{e_{1min}, e_{2min}\}$ is wrong. This value should be earlier in time.

Let us remark that, if there exist constraint violations, the involved life intervals can be considered as random intervals that are introduced in our system by chance. In the next subsection, we will explore some properties of the distribution of random intervals that can help to evaluate the capacity of a system to find contemporary constraint violations.

2.4 Properties of the Induced Contemporary Relation

We turn to study the properties of the induced contemporary relation $\mathcal{C}_E$ from a set of entities E . It can be trivially deduced from the definitions that contemporariness is *reflexive*, $(e_1, e_1) \in \mathcal{C}_E$; *symmetric*, $(e_1, e_2) \in \mathcal{C}_E \Leftrightarrow (e_2, e_1) \in \mathcal{C}_E$; and *non-transitive*, since $(e_1, e_2), (e_2, e_3) \in \mathcal{C}_E$ do not imply that $(e_1, e_3) \in \mathcal{C}_E$, with $e_1, e_2, e_3 \in E$. As a direct consequence, the cardinality of $\mathcal{C}_E$ is bounded by the number of entities in E and its square, $|E| \leq \mathcal{C}_E \leq |E|^2$.

The relation $\mathcal{C}_E$ induces what is called an *interval graph* [6]. The vertexes of this graph are the entities of the set E , and there exists

an edge between $e_i, e_j \in E$ when the corresponding life intervals intersect, $\mathcal{I}_{e_i} \cap \mathcal{I}_{e_j} \neq \emptyset$. If the endpoints of the life intervals are random variables, then the interval graph is said to be a *random interval graph*.

Several models to construct random interval graphs have been proposed and studied. In [14], the endpoints of a set with n intervals are constructed using $2n$ independent and identically distributed (i.i.d.) random variables. Under these assumptions, it can be proved [11] that the probability that two intervals intersect is $2/3$, with disregard of the distribution of the random variables. It can be proved as well that the expected number of intersections is $n(n-1)/3$, and so, the expected number of pairs belonging to $\mathcal{C}_E$ is $\mathbb{E}[|\mathcal{C}_E|] = (2|E|^2 + |E|)/3$.

Although contemporariness is not transitive, in the presence of random data, transitivity occurs with high probability. With random $\mathcal{I}_{e_1}, \mathcal{I}_{e_2}, \mathcal{I}_{e_3}$, probability of e_1 and e_3 being contemporary, given that $(e_1, e_2), (e_2, e_3) \in \mathcal{C}_E$, is

$$P((e_1, e_3) \in \mathcal{C}_E \mid (e_1, e_2), (e_2, e_3) \in \mathcal{C}_E) = \frac{9}{11}.$$

Hence, the graphs tend to be dense. Moreover, it is proven in [14] that these kinds of random interval graphs are dense, in the sense that, as the number of intervals — that is, the number of entities, $|E|$ — tends to infinity, the probability that the number of intersections is equal to $n^2/3 + o(n^2)$ tends to one. So, the cardinality of $\mathcal{C}_E$, as the number of entities grows, is likely to be close to the expected value $\mathbb{E}[|\mathcal{C}_E|] = (2|E|^2 + |E|)/3$.

Another model to construct random interval graphs is presented in [15]. We use a generalization of this model. It is considered that the midpoints of the intervals are i.i.d. random variables and the length of the intervals are also, but possibly different, i.i.d. random variables. As a consequence, the following result can be noted.

THEOREM 2.4. *Let $O = \{O_1, \dots, O_n\}$ and $W = \{W_1, \dots, W_n\}$, $W_i \geq 0$, be i.i.d. sets of random variables following the probability distributions F_O and F_W , respectively. If E is a set of n entities whose life intervals come from $[O_i - W_i, O_i + W_i]$ and p is the probability that two of those intervals intersect, then*

$$\mathbb{E}[|\mathcal{C}_E|] = |E|(|E| - 1)p + |E|.$$

This result shows that, if we want to calculate the expected number of pairs in the induced contemporary relation $\mathcal{C}_E$, it is only necessary to calculate the probability that two intervals, $[o_x - w_x, o_x + w_x]$ and $[o_y - w_y, o_y + w_y]$, intersect,

$$p = \int_{\mathbb{R}} P(|o_x - o_y| \leq t \mid w_x + w_y = t) d(P(w_x + w_y \leq t)). \quad (1)$$

Depending on the underlying distributions F_O and F_W , this integral has an exact solution. But we can always estimate the value p and, therefore, the expected value of the cardinality of $\mathcal{C}_E$ by sampling the endpoints of just two intervals. Hence, to estimate $\mathbb{E}[|\mathcal{C}_E|]$, we can always consider that we have a set of two entities, regardless of the real number of entities in E .

3 TEMPORAL FALSIFIABILITY

In this Section, we measure the capacity of an information system to find temporal constraint violations. It is easy to prove that the following conditions maximize the potential that a contemporary constraint has in order to detect wrong data:

- All the start and the end dates are defined, so $\bigcup \mathcal{I}_E \subseteq \mathbb{R}$.
- The life intervals of the entities, $\mathcal{I}_E$, are very short in relation to the range of accepted values.
- The start or the end dates are uniformly distributed across the entire range of accepted values.
- Each entity is related to all the others through the studied contemporary relationship.

To estimate, depending on some of these points, how useful a contemporary constraint can be when applied to a certain data set, we will introduce a metric named *temporal falsifiability*.

Definition 3.1. Given a non-empty set of entities E , the *temporal falsifiability* of E , which is denoted by f_E , is defined as the proportion of non-contemporary entity pairs in $E \times E$ over the total number of entity pairs in $E \times E$,

$$f_E := 1 - \frac{|\mathcal{C}_E|}{|E|^2}.$$

Let us note that, if $f_E = 0$, then all the possible entity pairs in $E \times E$ are contemporary, i.e. for any $e_1, e_2 \in E$, $\mathcal{I}_{e_1} \cap \mathcal{I}_{e_2} \neq \emptyset$. In this case, it will be useless to apply a contemporary constraint to some R_E , given that none of the wrong data would be detected. On the contrary, if $f_E = 1 - \frac{1}{|E|}$, then any violation of a contemporary constraint would be detected, as every non-reflexive entity pair in R_E would be inconsistent. It must be noted that always $0 \leq f_E \leq 1 - \frac{1}{|E|}$ due to the fact that $|E| \leq |\mathcal{C}_E| \leq |E|^2$. Hence, the temporal falsifiability captures the idea that, the less intersections between life intervals, the easier to detect a temporal constraint violation of any contemporary relation R_E .

A problem appears with the definition of temporal falsifiability. If the set of entities E is very large, it is needed to find the number of intersections of the life intervals to calculate f_E . This calculation can be unfeasible depending on the number of entities. As far as we know, the best algorithm to count the number of intersections of a set of intervals has time complexity $O(n \log(n))$, but it is needed to sort the endpoints, and therefore, the space complexity is $O(n)$. Of course, the best algorithm with constant space complexity has time complexity $O(n^2)$. So, a suitable estimation is required. In the next section, we will give two approaches to properly estimate the temporal falsifiability of a set of entities.

4 EXPERIMENTS

In this section, we begin by presenting and evaluating two methods to estimate the temporal falsifiability of a set of entities. After that, we focus on evaluate the compliance of Wikidata.

4.1 Estimation of the Temporal Falsifiability

In order to estimate the temporal falsifiability two approaches are considered: 1) estimation by modelling and 2) estimation by sampling.

4.1.1 Estimation by modeling. The first method to estimate the temporal falsifiability is based on a prior knowledge about the distribution of the life intervals of a set of entities E . If the probability distributions F_o and F_w of the mid points and length of the life intervals, respectively, are known, then, we can directly estimate

the temporal falsifiability of E , $\hat{f}_E$, by Theorem 2.4,

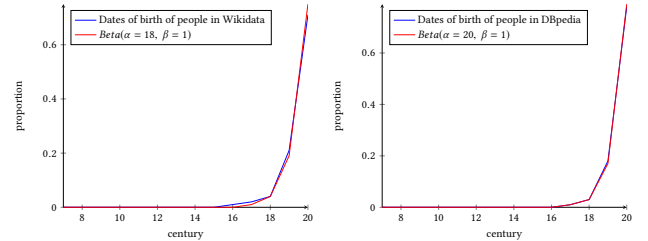
$$\hat{f}_E = \frac{(n-1)(1-p)}{n}, \quad (2)$$

where $n = |E|$ and p is the value of the integral of Equation 1.

To show the validity of this method, we have conducted experiments using as set of entities the people that appear in Wikidata and DBpedia from the 7th century to the 20th century. Firstly, we need to construct a model for the life intervals distribution, i.e. we need to find probability distributions F_o and F_w that approximate as well as possible to the real data. To achieve this goal, we consider:

- Birth dates are determined by a beta distribution with $\alpha \geq 1$ and $\beta = 1$. In many data sources with historical data, such as Wikidata and DBpedia, most registered people have recent birth dates and there are less people with old birth dates (Fig. 1). The expected trend of any organizational database is storing more recent information faster as the organization grows. Over time, more information is generated and stored about every living person [17, 18] and the human population is not expected to stop growing in the coming decades [8].
- The Gompertz distribution is commonly used in social sciences to model the life span of people. Death dates are computed by adding the birth dates to the lengths of the life intervals that are suggested by a Gompertz distribution with a shape of $\frac{\log(2)}{8}$ and a rate of $\frac{1}{5000}$ [5], but only when the resulting value is not later than the latest date allowed by the range constraint. Otherwise, the value is considered $+\infty$.

Figure 1: Dates of birth of people in Wikidata (left) and DBpedia (right) each fit a beta distribution with $\beta = 1$ from the 7th to the 20th century. Data retrieved from <https://tinyurl.com/yargwft2> and <https://tinyurl.com/y7jc9efn>.



Under these assumptions, we model the birth dates for Wikidata by a distribution $\text{Beta}(18, 1)$, and for DBpedia by $\text{Beta}(20, 1)$ (see Fig. 1). The lengths of the life intervals in both cases is modeled by $\text{Gompertz}(\frac{\log(2)}{8}, \frac{1}{5000})$.

	f_E	$\hat{f}_E$	$\hat{f}_E - f_E$
Wikidata	0.37	0.43	0.05
DBpedia	0.38	0.39	0.01

Table 1: Real temporal falsifiability f_E , its estimated value $\hat{f}_E$, and the difference $\hat{f}_E - f_E$ for the set of people in Wikidata and DBpedia from the 7th century to the 20th century.

Once we have obtained the model, we can estimate the temporal falsifiability. In Table 1, the obtained results can be seen. The estimated values are similar to the real ones, obtaining a value for the DBpedia almost equal to the real one. The reason to obtain this result for DBpedia is that the beta distribution for its birth dates reflects the reality better than for Wikidata.

Temporal Falsifiability of People. As an example of how this method can be applied to a specific domain (i.e., a set of entities), we calculate the temporal falsifiability by means of the model suggested by the people in Wikidata and DBpedia analyzed in the above point. People are a common kind of entities in many databases and they participate in a large number of contemporary relationships. The temporal falsifiability of people is going to be quantified so that it can be consulted depending on two parameters:

- The earliest acceptable year for birth and death dates until the present time, that is to say, the length of the range of accepted values for birth and death dates.
- The degree to which life intervals are concentrated in the most recent years instead of being evenly distributed across the entire range of accepted values.

With the described data sets, different values of $f_E = 1 - \frac{|C_E|}{|E|^2}$ are calculated. These values, which are exposed in Table 2, can be used to determine if, given a certain data source with an approximate α , and a length of the range of accepted dates in years, the decision of applying or implementing contemporary constraints should be made. It is important to remind that, if some applicable birth or death dates are not defined, then f_E can be less than expected.

4.1.2 Estimation by sampling. The main problem with the first estimation method explained above is that we must have a model for the life intervals distribution of our domain. Depending on the data source, finding such model can be a very difficult or impossible task.

In the absence of a complete knowledge of the distribution of life intervals, we can use a simple algorithm to estimate the temporal falsifiability. Let us remind that, in order to calculate the temporal falsifiability, we can find the probability p that two life intervals from E intersect, and apply Equation 2. Algorithm 1, is considered to return the estimated temporal falsifiability in this case.

```

1  $c := 0$ 
2 repeat  $|E|$  times
3   pick_random  $e_1, e_2$  from  $E$  where  $e_1 \neq e_2$ 
4   if  $\mathcal{I}_{e_1} \cap \mathcal{I}_{e_2} \neq \emptyset$  then  $c++$ 
5 end
6  $\hat{f}_E := \frac{(|E| - 1)(|E| - c)}{|E|^2}$ 
7 return  $\hat{f}_E$ 

```

Algorithm 1: Algorithm to estimate the temporal falsifiability $\hat{f}_E$ of E

We calculate the estimated and real temporal falsifiability for some data sources from the Web to validate Algorithm 1. In particular, we have used Wikidata, DBpedia, dati.camera.it, and Sabi under the domains of people, companies, and political parties. The available birth and death dates have been used when the domain was

“people”, and founding and dissolution dates for companies and political parties. For Wikidata, DBpedia and dati.camera.it, the SPARQL endpoints were used to retrieve data: <https://query.wikidata.org/sparql>, <https://dbpedia.org/sparql> and <http://data.camera.it/sparql>, while for Sabi, the data has been downloaded through its web interface.

We have run 20 times the Algorithm 1 for each domain. After that, the average and the variance of the 20 obtained values were computed in order to estimated a value for $\hat{f}_E$ (see Table 3). As it can be noted from Table 3, the real and estimated values are equal up to two decimals with a variance always lower than $5 \cdot 10^{-4}$. We also include the temporal falsifiability when the endpoints of the life intervals follow a random variable. This value can be used as a base ground of how far our information is from a set of random intervals.

Hence, it can be concluded from Table 3 that the best domain to apply the contemporary constraint is the selected sovereign states from Wikidata. We would like to stress that a high value of the temporal falsifiability do not imply that we can expect to have more inconsistencies; it means that the possible inconsistencies are more likely to be detected.

4.2 Compliance of Wikidata

Wikidata is an online, freely usable and editable knowledge base created in 2012 with the aim of centralizing and providing data about the world in several languages [12] to the Wikimedia projects (including Wikipedia) and other services and applications. Anyone can edit Wikidata, so it is particularly important to define and implement some mechanisms to ensure its data consistency.

Having studied and stated the fundamentals of contemporary constraints, and with the aim of implementing contemporary constraints in Wikidata, we have identified the most relevant Wikidata properties to which a contemporary constraint should be applied to actively ensure data quality. The contemporary properties with the highest rate of violations are presented in Table 4 with their relevant data⁴.

From our study of the content of Wikidata, we can claim the following conclusions:

- The contemporary property *country of citizenship* (P27) concentrates the highest rate of violations. This is caused mainly by:
 - (1) users have difficulties to realize that historical people do not correspond to current countries but to historical countries. For instance, *Marie-Rosalie Cadron-Jetté*, <http://www.wikidata.org/entity/Q628365>, a nun and midwife who died in 1864 is considered to be a citizen of Canada, <http://www.wikidata.org/entity/Q16>. However, the start date of Canada is said to be 1867. Both dates can be considered correct, so the value defined as *country of citizenship* is wrong and the right one could correspond to the United Kingdom.
 - (2) the definition and limits of many administrative territorial entities in Wikidata are inconsistent with each other. While some of them consider the date of the first human

⁴The data were retrieved from <https://tinyurl.com/y82o47rr> and <https://dumps.wikimedia.org/wikidatawiki/entities/> in April 2018.

Table 2: Values of the temporal falsifiability, f_E , of different sets of average people given the α of the beta distribution with $\beta = 1$, that birth dates follow, given the length of the range of accepted dates in years. Values $f_E < \mathbb{E}[f_E] = 1/3$ are red colored.

years	α of the beta distribution with $\beta = 1$											
	1	2	3	4	5	7	10	13	18	25	35	48
7000	0.98	0.98	0.97	0.96	0.95	0.93	0.91	0.89	0.84	0.80	0.73	0.64
6000	0.98	0.97	0.96	0.95	0.95	0.93	0.89	0.86	0.82	0.77	0.69	0.59
5000	0.98	0.97	0.96	0.94	0.94	0.90	0.87	0.84	0.79	0.72	0.64	0.55
4000	0.97	0.96	0.95	0.93	0.91	0.89	0.84	0.81	0.75	0.67	0.58	0.48
3000	0.96	0.94	0.93	0.90	0.89	0.86	0.79	0.75	0.68	0.58	0.46	0.35
2000	0.94	0.92	0.89	0.86	0.84	0.79	0.71	0.65	0.55	0.44	0.34	0.23
1500	0.92	0.90	0.86	0.82	0.79	0.72	0.63	0.57	0.45	0.35	0.22	0.14
1000	0.88	0.84	0.79	0.73	0.70	0.61	0.50	0.41	0.31	0.20	0.13	0.05
800	0.85	0.80	0.75	0.68	0.64	0.54	0.43	0.34	0.23	0.13	0.08	0.03
600	0.80	0.74	0.67	0.59	0.54	0.44	0.32	0.24	0.15	0.08	0.04	0.01
500	0.76	0.70	0.62	0.55	0.48	0.37	0.26	0.18	0.10	0.05	0.02	0.01
400	0.71	0.64	0.54	0.46	0.40	0.28	0.18	0.11	0.06	0.03	0.01	0.01
350	0.67	0.59	0.49	0.41	0.34	0.23	0.14	0.09	0.04	0.02	0.01	0.00
300	0.62	0.53	0.43	0.35	0.28	0.18	0.10	0.06	0.02	0.01	0.00	0.00
250	0.57	0.46	0.35	0.28	0.22	0.13	0.07	0.04	0.01	0.01	0.00	0.00
200	0.48	0.36	0.25	0.19	0.14	0.07	0.03	0.02	0.01	0.00	0.00	0.00
150	0.34	0.23	0.15	0.09	0.07	0.03	0.01	0.01	0.00	0.00	0.00	0.00
100	0.14	0.08	0.05	0.03	0.02	0.01	0.00	0.00	0.00	0.00	0.00	0.00
75	0.06	0.03	0.02	0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
50	0.01	0.01	0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table 3: Data sources and the studied domain for Wikidata, DBpedia, dati.camera.it and Sabi, sorted by temporal falsifiability.

Data source	Domain E	f_E	$\hat{f}_E$	$\text{Var}[\hat{f}_E]$
wikidata.org	243 sovereign states having their dates of creation and articles on English Wikipedia defined	0.54	0.54	$4.99 \cdot 10^{-4}$
dbpedia.org	2100 people having their dates of birth and death defined	0.47	0.47	$1.22 \cdot 10^{-4}$
wikidata.org	500 000 people having their dates of birth defined	0.37	0.37	$3.67 \cdot 10^{-8}$
	Random data	0.33	n/a	n/a
dati.camera.it	10 000 deputies having their dates of birth defined	0.26	0.26	$1.18 \cdot 10^{-5}$
wikidata.org	9414 political parties having their dates of creation defined	0.25	0.25	$1.06 \cdot 10^{-5}$
sabi.bvdinfo.com	249 800 companies having their dates of creation defined	0.11	0.11	$2.50 \cdot 10^{-7}$

settlement in that location as the start date of a current entity, others consider as its start date the moment when the current political regime was established. Unless some common guidelines are agreed on this topic, the use of this property will not be consistent and the rate of violations will not be significantly reduced. Nevertheless, it is important to point out that the data about countries are consistent thanks to their high impact on the entire project of Wikidata: despite the fact that there are only a

few countries, these countries are linked from millions of other entities.

- Some entities are considered different when they are re-opened or refounded after a period of inactivity. On the other hand, some others are considered to be the same under these circumstances. As consequence, some users link entities following a criterion despite the fact that the entities being linked were conceived following a different one. In some

Table 4: Some Wikidata properties that should be contemporary. Their IDs, shortened labels, current violation rates, $|[R_E]_v| / |R_E|$, and rates of entity pairs $(e_1, e_2) \in R_E$ with eternal subjects, $\mathcal{I}_{e_1} = (-\infty, +\infty)$, and objects, $\mathcal{I}_{e_2} = (-\infty, +\infty)$, that is to say, entities without start and end dates, are shown.

ID	short label	viol.	et. e_1	et. e_2
P27	... citizenship	4.25%	16.3%	16.3%
P108	employer	1.28%	8.3%	7.0%
P937	work location	1.10%	7.4%	32.8%
P20	place of death	0.99%	0.5%	49.6%
P607	conflict	0.99%	17.1%	7.8%
P1066	student of	0.92%	3.2%	2.7%
P38	currency	0.92%	28.8%	74.0%
P157	killed by	0.92%	43.9%	47.8%
P802	student	0.86%	3.2%	3.9%
P1037	manager/director	0.84%	17.2%	18.0%
P2632	place of detention	0.83%	2.9%	30.7%
P169	CEO	0.72%	10.2%	23.3%
P1830	owner of	0.71%	21.6%	38.9%
P26	spouse	0.60%	11.7%	12.1%
P69	educated at	0.54%	8.6%	4.5%
P19	place of birth	0.54%	2.2%	58.8%
P101	field of work	0.51%	11.1%	96.5%
P551	residence	0.42%	38.1%	60.3%
P749	parent org.	0.40%	32.9%	21.5%
P451	partner	0.36%	24.8%	24.9%
P463	member of	0.31%	17.9%	25.7%
P706	located on ...	0.29%	89.9%	91.9%
P123	publisher	0.27%	83.9%	11.5%
P102	member of ... party	0.25%	6.2%	2.98%
P488	chairperson	0.25%	14.1%	9.33%
P35	head of state	0.24%	29.1%	9.38%
P1923	participating teams	0.23%	67.3%	10.3%
P47	shares border with	0.22%	90.5%	90.4%
P118	league	0.21%	10.0%	5.67%
P286	head coach	0.20%	4.7%	12.7%
P140	religion	0.20%	21.1%	37.8%
...	...	...	...	...

of these cases, a violation of the contemporary constraint arises.

- Many start and end dates are missing when they are not known with an exact precision of years, months or days. However, Wikidata allows to define dates with any degree of precision. Dates with a precision of decades, centuries or millenia should be defined when applicable, especially for the most linked entities. This would increase both the completeness of Wikidata and the efficacy of contemporary constraints.

Finally, we would like to emphasize that just considering the properties⁵ *country of citizenship* (P27), *employer* (P108), *work location* (P937), *place of death* (P20), and *conflict* (P607), which are

⁵Every property in Wikidata is identified by a URI that respects the pattern <http://www.wikidata.org/entity/<ID>>, where <ID> is the character P followed by an integer.

those with the highest rate of violations (see Table 4), the number of violations is around 130 000. Under this circumstances, we consider that a semi-automated tool to fix the inconsistencies or highlight them could be positive.

5 RELATED WORK

The exploitation of digital data, both freely available on the Web and private, usually requires a data cleaning process in order to remove inconsistencies in the input data to be considered and to transform the data into specific formats. These tasks consume, on average, more than 80% of the time and budget of all exploitation work flow [23]. So, in the last decades, there has been an increasing interest in developing techniques and tools to measure and improve the data quality and to detect data inconsistencies. Thus, in [23], a distributed stream data cleaning system, called Bleach, is presented. The main difference of Bleach with respect to previous tools is that it does not rely on batch processing to improve data quality, but data is cleaned incrementally in real-time. The mechanisms to check the consistency of the data in this paper can be used also incrementally in real-time. Therefore, they could be integrated in Bleach, given that, as far as we know, it does not consider any constraint related to the coexistence of entities in its cleaning processes.

Temporal constraint satisfaction and deciding consistency tasks have been widely studied from different perspectives that range from data management to computational intelligence and knowledge representation [16]. However, to the best of our knowledge, there is no previous work about random interval graphs that formalizes and analyzes the data consistency of Linked Data sources and the Semantic Web. Some recent works focus on maintaining integrity constraints in the translation of relational data sources to semantic web data sources, but they do not deal with constraints related to temporal data or coexistence [4]. Thus, inconsistency detection tools, such as the reasoners commonly used in the semantic Web community, cannot detect inconsistent data by considering the contemporary constraint.

Related to our work, but with different goals, fact-checking is being an active matter of study; in particular, the temporal scope of facts⁶. There are different approaches to validate the scope of temporal information. In [22] the authors present Timely YAGO, that uses regular expressions and rule-based techniques to find and validate the time range of facts on Wikipedia infoboxes. In this context, PRAVDA [21] uses textual web sources to augment a fact with a time annotation. The authors use a graph-based semi-supervised method, *label propagation*, to obtain the time points of a fact. CoTS system [19] uses document creation times (DCT) to temporally scope facts by querying to Google Books and Gigaword datasets to collect documents, by using the DCTs of the retrieved documents and a Maximum Entropy classifier to compute a probability that a fact was true at a given a time and, finally, by applying temporal constraints and an Integer Linear Program to predict which facts are true at a given time. Another system that not only relies on temporal information of a fact, but also on the trustworthiness of the fact, is DeFacto [7]. This system uses textual information in the Web to obtain facts given as RDF triples. Then it is used a supervised method to obtain both if the fact is true and its scope.

⁶E.g., checking if Clinton was the president of the United States from 1993 to 2001.

Finally, we would like to remark that there exists a great amount of works about methodologies to evaluate the data quality of different data sources [24]. The dimensions adopted in the literature to evaluate the quality of the data in the Linked Open Data domain are typically grouped in categories in the following categories: Intrinsic, Contextual, Representational and Accessibility. In this context, the contemporary constraint is considered to be a mechanism useful to evaluate dimensions in the category *Intrinsic*, in particular to evaluate the accuracy and the consistency of a data source.

6 CONCLUSIONS AND FUTURE WORK

As data grows in information systems, there is a need to verify and detect the veracity and consistency of such information. In this paper, we have focused on temporal data to improve the quality of the data. In particular:

- We have defined and formalized the concept of contemporary relationship and contemporary constraint violation, concepts that can be used to detect temporal inconsistencies in an information system.
- We have also defined a metric, temporal falsifiability, to measure the capacity of an information system to detect contemporary constraint violations.
- We have shown two methods to estimate the temporal falsifiability of an information system, and we have calculated it for a set of real data sources.
- Finally, we have calculated and analyzed the contemporary constraints violations for some contemporary relations in Wikidata.

In summary, we have shown that the contemporary relations are a useful to find inconsistent data in an information system.

As future work, we would like to enhance the definition of temporal falsifiability to consider not only the life intervals of entities, but the graph of the defined contemporary relations. Another interesting problem that we are currently analyzing is how to discover inconsistencies without exploring all the data; i.e., applying heuristics to find entities that are more likely to participate in a contemporary constraint violation. In order to do this, we are exploring the use of the theory of interval graphs. Notice that, if there are no contemporary violations, then the graph induced by the contemporary relations must be a subgraph of the graph induced by the life intervals. It is known that the subgraph isomorphism problem for interval graphs can be solved in polynomial time, so we will investigate heuristics to detect graph dissimilarities that, in the context of this problem, are wrong data.

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REFERENCES

- [1] James F. Allen. 1983. Maintaining Knowledge About Temporal Intervals. *Commun. ACM* 26, 11 (Nov. 1983), 832–843. <https://doi.org/10.1145/182.358434>
- [2] S. Auer, C. Bizer, G. Kobilarov, J. Lehmann, R. Cyganiak, and Z. Ives. [n. d.]. DBpedia: A Nucleus for a Web of Open Data, In 6th International Semantic Web Conference, (ISWC 2007). *The Semantic Web*, 722–735. https://doi.org/10.1007/978-3-540-76298-0_52
- [3] Franz Baader, Diego Calvanese, Deborah L. McGuinness, Daniele Nardi, and Peter F. Patel-Schneider. 2010. *The Description Logic Handbook: Theory, Implementation and Applications* (2 ed.). Cambridge University Press, New York, NY, USA.
- [4] Ming Fang. 2013. *Maintaining Integrity Constraints in Semantic Web (in particular, Chapter 3 and 4, pages 31–49)*. Ph.D. Dissertation. Georgia State University, USA. http://scholarworks.gsu.edu/cs_diss/73
- [5] CE Finch, MC Pike, and M Witten. 1990. Slow mortality rate accelerations during aging in some animals approximate that of humans. *Science* 249, 4971 (1990), 902–905. <https://doi.org/10.1126/science.2392680> arXiv:<http://science.sciencemag.org/content/249/4971/902.full.pdf>
- [6] Peter C. Fishburn. 1985. *Interval Orders and Interval Graphs: A Study of Partially Ordered Sets*. New York: Wiley.
- [7] Daniel Gerber, Diego Esteves, Jens Lehmann, Lorenz BÄßmann, Ricardo Usbeck, Axel-Cyrille Ngonga Ngomo, and RenÄ Speck. 2015. DeFacto-Temporal and multilingual Deep Fact Validation. *Journal of Web Semantics* 35 (2015), 85 – 101. <https://doi.org/10.1016/j.websem.2015.08.001> Machine Learning and Data Mining for the Semantic Web (MLDMSW).
- [8] Patrick Gerland, Adrian E. Raftery, Hana Ševčíková, Nan Li, Danan Gu, Thomas Spoorenberg, Leontine Alkema, Bailey K. Fosdick, Jennifer Chunn, Nevena Lalic, Guiomar Bay, Thomas Buettner, Gerhard K. Heilig, and John Wilmoth. 2014. World population stabilization unlikely this century. *Science* 346, 6206 (2014), 234–237. <https://doi.org/10.1126/science.1257469> arXiv:<http://science.sciencemag.org/content/346/6206/234.full.pdf>
- [9] Valentin Goranko and Antony Galton. 2015. Temporal Logic. In *The Stanford Encyclopedia of Philosophy*, Edward N. Zalta (Ed.). Metaphysics Research Lab, Stanford University.
- [10] Stefan Heindorf, Martin Potthast, Benno Stein, and Gregor Engels. 2015. Towards Vandalism Detection in Knowledge Bases: Corpus Construction and Analysis. In *38th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '15)*. ACM, NY, USA, 831–834. <https://doi.org/10.1145/2766462.2767804>
- [11] Joyce Justicz, Edward R. Scheinerman, and Peter M. Winkler. 1990. Random Intervals. *Am. Math. Monthly* 97, 10 (Nov. 1990), 881–889. <https://doi.org/10.2307/2324324>
- [12] Lucie-Aimée Kaffee, Alessandro Piscopo, Pavlos Vougiouklis, Elena Simperl, Leslie Carr, and Lydia Pintscher. 2017. A Glimpse into Babel: An Analysis of Multilinguality in Wikidata. In *13th International Symposium on Open Collaboration (OpenSym '17)*. ACM, NY, USA, Article 14, 5 pages. <https://doi.org/10.1145/3125433.3125465>
- [13] Andrei Krokhin, Peter Jeavons, and Peter Jonsson. 2003. Reasoning About Temporal Relations: The Tractable Subalgebras of Allen's Interval Algebra. *J. ACM* 50, 5 (Sept. 2003), 591–640. <https://doi.org/10.1145/876638.876639>
- [14] E. R. Scheinerman. 1988. Random interval graphs. *Combinatorica* 8, 4 (01 Dec 1988), 357–371. <https://doi.org/10.1007/BF02189092>
- [15] Edward R. Scheinerman. 1990. An evolution of interval graphs. *Discrete Mathematics* 82, 3 (1990), 287 – 302. [https://doi.org/10.1016/0012-365X\(90\)90206-W](https://doi.org/10.1016/0012-365X(90)90206-W)
- [16] Eddie Schwalb and Lluís Vila. 1998. Temporal Constraints: A Survey. *Constraints* 3, 2/3 (June 1998), 129–149. <https://doi.org/10.1023/A:1009717525330>
- [17] SINTEF. 2013. Big Data, for better or worse: 90% of world's data generated over last two years. <https://www.sciencedaily.com/releases/2013/05/130522085217.htm>; accessed April 2018.
- [18] Latanya Sweeney. 2001. Information explosion. *Confidentiality, disclosure, and data access: Theory and practical applications for statistical agencies* (2001), 43–74.
- [19] Partha Pratim Talukdar, Derry Wijaya, and Tom Mitchell. 2012. Coupled Temporal Scoping of Relational Facts. In *Fifth ACM International Conference on Web Search and Data Mining (WSDM '12)*. ACM, NY, USA, 73–82. <https://doi.org/10.1145/2124295.2124307>
- [20] Denny Vrandečić. 2012. Wikidata: A New Platform for Collaborative Data Collection. In *21st International Conference on World Wide Web (WWW '12 Companion)*. ACM, New York, NY, USA, 1063–1064. <https://doi.org/10.1145/2187980.2188242>
- [21] Yafang Wang, Bin Yang, Lizhen Qu, Marc Spaniol, and Gerhard Weikum. 2011. Harvesting Facts from Textual Web Sources by Constrained Label Propagation. In *20th ACM International Conference on Information and Knowledge Management (CIKM '11)*. ACM, NY, USA, 837–846. <https://doi.org/10.1145/2063576.2063698>
- [22] Yafang Wang, Mingjie Zhu, Lizhen Qu, Marc Spaniol, and Gerhard Weikum. 2010. Timely YAGO: Harvesting, Querying, and Visualizing Temporal Knowledge from Wikipedia. In *13th International Conference on Extending Database Technology (EDBT '10)*. ACM, NY, USA, 697–700. <https://doi.org/10.1145/1739041.1739130>
- [23] Tian Yongchao. 2017. *Accelerating Data Preparation for Big Data Analytics (in particular, Chapter 1. Introduction, pages 1–16)*. Ph.D. Dissertation. Paris Tech Paris Institut des sciences et technologies, Paris, France.
- [24] Amrapali Zaveri, Anisa Rula, Andrea Maurino, Ricardo Pietrobon, Jens Lehmann, and Sören Auer. 2015. Quality Assessment for Linked Data: A Survey. *Semantic Web Journal* 7, 1 (2015), 63–93. <http://www.semantic-web-journal.net/content/quality-assessment-linked-data-survey>